

Required Assignment 10.1: Satellite Data and Fintech Innovation for Financial Inclusion

Name: Raja Shekar Bollam

Based on the case study “Launching into Space: Using Satellite Imagery in Financial Services” (Apollo Agriculture and Harvesting Inc.), Partnership for Finance in a Digital Africa / Caribou Digital, 2018.

1. Satellite Data in Agriculture and Financial Inclusion

Earth observation data lets these FinTechs “see and try to understand what is going on from anywhere and even look backward in time” (Apollo), which is vital where farmers lack credit histories. Satellites capture what traditional means cannot — crop yields, crop mapping, geotagging farmland to specific farmers, farm diversification, and planting cycles — enabling forecasts of revenue, repayment deficits, and income timing. Free government imagery (Europe’s Copernicus Sentinel, the USGS Earth Explorer) at 10-metre resolution, refreshed every 6–12 days, lets Apollo and Harvesting geotag farms down to half an acre; Sentinel’s radar even images through clouds. For financial inclusion, this turns “thin-file” smallholders — who make up 70% of Kenya’s agricultural production yet lack affordable credit — into scorable customers by combining imagery with demographic, financial, agronomic, and psychometric data in machine-learning credit models. It also cuts the operational and logistical cost of reaching remote rural clients, making lending viable. Harvesting’s Farmland Monitoring Tool tracks loan performance across the cropping cycle and alerts banks to likely defaults, so they can extend or resize loans — protecting both lender and farmer.

2. B2B vs. B2C Model for Harvesting Inc.

Harvesting chose B2B — selling AI-driven credit-scoring models and its Farmland Monitoring Tool to agri-lending banks and microfinance institutions — because it believed it could “reach scale by working with financial institutions while monetizing their services as a for-profit company.” Partnering with FSPs that already have a rural footprint lets Harvesting use technology and data to reduce the marginal cost of customer acquisition, earning upfront integration fees and per-farmer subscriptions without holding loans itself. A modified value-appropriation strategy targeting B2C was possible — going direct to farmers, as Apollo does — but the trade-offs are steep. Apollo’s own experience shows customer acquisition costs are “more challenging to recover... for farmers with low-value loans,” and a direct model would force Harvesting to absorb lending risk, field operations, and capital it currently avoids. B2C could capture more of the value chain and own the end-customer relationship, but at the cost of asset-light scalability. Conversely, the B2B trade-off is dependence on slow-moving institutions — Harvesting first had to streamline one MFI’s data systems before it would even consider satellite imagery — and managing clients’ unrealistic expectations of the technology.

3. Assessing Apollo and Harvesting’s Data-Driven Capabilities

Yes — both demonstrate the core capabilities of data-driven companies, though with real challenges. On talent, their data-science teams hold advanced degrees directly relevant to the problem — PhDs in Experimental Physics and Remote Sensing Science, and master’s degrees in engineering and Agricultural Engineering — spanning the four required skills (software engineering, ML/data science, remote sensing, and agricultural science), built in-house. On infrastructure, Harvesting built “highly scalable super computing infrastructure on cloud with 11,000 CPUs” to process imagery continuously. On data strategy, Apollo combined satellite imagery with psychometric, financial, and observational data, comparing predictive power to select the strongest combination, and automated its supply chain — acquisition, disbursement, extension, and collection — without field staff. Founder pedigree reinforces this: Apollo’s team came from One Acre Fund and The Climate Corporation. The challenges are equally clear — long annual data-collection cycles constrain the pace of learning (“which can kill you as a business”), training data is scarce or low quality in East Africa, and customer acquisition remains the hardest, costliest step. Both are genuinely capable, but still early in proving their models at scale.

4. Importance of Quality Training Data

High-quality training data is hard and expensive because agricultural data “often requires an entire growing season” to collect, so models improve only once a year, and in East Africa such data “either does not exist... or is low quality.” A good model needs diverse data over a long period and many customers, yet repayment behavior is driven by shifting factors like weather, demanding still more data. Experimentation overcomes this: the Fintechs “constantly experiment with data-driven models,” and Apollo deliberately went “out and just making loans to gather data even though it’s higher risk,” accepting that they “will fail a few times to get it right” (Harvesting). By feeding more data through — including satellite signals on how fields have performed — the models become more accurate and gain predictive power for next season’s crops. This is critical because predictability of the credit model is a core KPI: as Harvesting notes, “the more data that feeds through, the more accurate the models and the greater the predictive power” for repayment. Without good training data, neither company can prove satellite imagery’s value, price risk, or build sustainable, scalable lending.